

SATYABHAN UMED BHADORIYA

AI/LLM Systems Engineer | Autonomous Agents | Low-Latency RAG | Production Systems

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PROFESSIONAL SUMMARY

AI/LLM Systems Engineer with 2+ years building production autonomous AI agents, high-throughput RAG systems, and latency-optimized inference pipelines, plus 8+ years of enterprise engineering discipline across Linux/RHEL systems and STIG security compliance. Creator of **CLINIRISE** — a production multi-clinic WhatsApp CRM powered by Google Gemini, Qdrant, Redis, React 18 / TypeScript frontend, and GCP Cloud Run — solo-built end-to-end with **860+ automated tests** and currently in active pilot with 2 clinic clients. Applies typed decision models (**TypeSafe Jev**) for fast verification and guardrail enforcement in agentic workflows. Author of **300+ chapters** of open-source AI/ML, DevOps-Infra, and Linux-HPC-Security engineering curricula with CI-validated labs.

FEATURED PROJECT — CLINIRISE (PILOT TESTING, 2 CLINIC CLIENTS)

Autonomous Healthcare CRM & RAG System — solo-built, end-to-end

Live app → gen-lang-client-0155046437.web.app

- **End-to-End Autonomous Agent:** Architected a WhatsApp-based conversational CRM integrating WhatsApp Business API with Google Gemini LLM for multi-turn patient scheduling, clinic FAQs, and autonomous booking with 2-way Google Calendar OAuth sync.
- **Full-Stack Architecture:** React 18 + TypeScript + Vite frontend, Express / Node.js backend on **GCP Cloud Run**, Firestore NoSQL, Turborepo monorepo. **860+ automated tests** (Vitest unit + Cypress E2E + k6 load testing). BigQuery + Looker Studio per-clinic analytics dashboards.
- **High-Throughput RAG:** Hybrid retrieval via Qdrant vector search + two-tier Redis caching, targeting sub-2-second response latency. Semantic intent routing separates deterministic transactional actions from generative queries, reducing average LLM API token consumption by ~40%.
- **Production Reliability:** Multi-tier AI resiliency (Gemini Flash → Flash-Lite → template fallback), clinical guardrails with human handoff escalation, DPDP Act consent & PII redaction, GitHub Actions CI/CD with Workload Identity Federation (zero service-account keys), scale target of 1,000 clinics × 100 patients/day.

EXPERIENCE

Generative AI / Systems Engineer | [Aman Security India Pvt. Ltd.](#)

March 2024 – Present

Deployed on-site at CBSE Regional Office, Pune | Full-time day role, alongside independent AI/LLM product development

- Self-initiated and independently built **CLINIRISE**, a production LLM-powered multi-clinic WhatsApp CRM, outside core duties — applying Gen AI, Python, LLMs, RAG, Vertex AI, React, TypeScript — currently in pilot testing with real clinic clients.
- **Voice AI Calling Agent:** Engineered an ultra-low-latency inbound/outbound voice AI agent pipeline integrating the **Tabby** platform, ElevenLabs, Bolna AI, Groq (fast LLM inference), and Azure STT for automated STT/TTS conversational flows.
- **LLM Evals & Cost Optimization:** Benchmarked multiple STT, LLM inference, and TTS platforms against Time-to-First-Token (TTFT), error drift, and token unit costs to select the most cost-effective production voice stack. Applied **TypeSafe Jev** typed decision model (Choice/Score/Boolean) for fast, low-cost verification and guardrail enforcement in agentic workflows, in place of full LLM calls for deterministic checks.
- **AI-Assisted Engineering & Infrastructure:** Used Claude Code and the Antigravity IDE as an AI-assisted coding workflow with GitHub Actions CI/CD. Monitored infrastructure health via Prometheus & Grafana; managed virtualized production environments with VMware; issue resolution via ServiceNow ticketing and Jira task tracking.

Linux Administrator | [Oceanic EOPS LLP](#)

2023 – 2024

Pune, India (remote support for client sites)

- Managed and maintained RHEL 7 production servers remotely for client sites, handling day-to-day server administration and issue resolution.
- Applied **DISA STIG security hardening** controls and compliance auditing via OpenSCAP; configured SELinux policies and Firefox browser configuration restrictions to establish high-assurance runtime boundaries.
- Tracked and resolved client support tickets via Skype; monitored server health and performance using Grafana dashboards.

Application Engineer | [Precision Automation](#)

2018 – 2023

Pune, India

- Delivered end-to-end automation engineering on-site across roughly 15 projects spanning assembly lines, packaging, and material handling systems — from raw material input through finished assembly.
- Designed and programmed HMI interfaces and PLC logic using GX Works (Mitsubishi) for industrial automation systems; used SolidWorks for mechanical design and CAD work. Deterministic state-machine control with 99.8% operational uptime.

SKILLS

AI & Agents:	Generative AI (Gen AI), Autonomous Agents, Tool/Function Calling, RAG Pipelines, Context Engineering, LLM Evals & Benchmarks, TypeSafe Jev (Typed Decision & Guardrail Model), Prompt Engineering, LangChain, ReAct Agents
Models & Inference:	Python, Google Gemini, Anthropic Claude, OpenAI, Vertex AI, Groq (Fast Inference), Ollama, PyTorch, Hugging Face Transformers, Scikit-learn, LoRA/DPO Fine-Tuning, INT8 Quantization, KV-Cache, Continuous Batching, vLLM, KServe
Voice AI:	ElevenLabs, Bolna AI, Azure STT, Tabby, STT/TTS Pipeline Design
Full-Stack:	React 18, TypeScript, Vite, Node.js, Express, Turborepo, FastAPI, Tailwind CSS, Razorpay, Google OAuth
Vector & Data:	Qdrant (Vector DB), Redis / Upstash (Caching & Sessions), Firestore (NoSQL), PostgreSQL, BigQuery, Looker Studio
Infra & DevOps:	Docker, Kubernetes, Terraform, Helm, GitOps, GCP Cloud Run, Firebase Hosting, GitHub Actions CI/CD, Linux / RHEL 7, OpenSCAP, STIG Hardening, Prometheus, Grafana, OpenTelemetry, VMware, Workload Identity Federation
Testing & SRE:	Vitest (860+ tests), Cypress E2E, k6 Load Testing, SLI/SLO/Error Budgets, Chaos Engineering, ServiceNow, Jira, Git/GitHub

OPEN-SOURCE ENGINEERING & BENCHMARKING

Author & Maintainer — 300+ Chapter Engineering Curricula (3 Repos, CI-Validated)

satyabhan007.github.io/AI-ML

- **AI-ML** (github.com/satyabhan007/ai-lab): 9-course, 138-chapter AI/ML engineering curriculum built from scratch in pure Python — scalar autograd engine with topological backpropagation (micrograd), BPE tokenizer, scaled dot-product attention with causal masks, Word2Vec embeddings, RAG + ReAct agent loops, LoRA/DPO fine-tuning, INT8 quantization, KV-cache, continuous batching, vLLM/KServe/Triton model serving. **69/69 automated CI checks passing**.
- **DevOps-Infra** (github.com/satyabhan007/DevOps-Infra): Terraform IaC (modules, remote state, drift detection, policy-as-code), Docker multi-stage hardened builds, Kubernetes (10 validated manifests, CRDs/operators, Helm, GitOps, service mesh, autoscaling), SRE playbooks (SLI/SLO/error budgets, golden signals, incident runbooks). 4-gate CI pipeline (validate + terraform + docker + kubeconform).
- **Linux-HPC-Security** (github.com/satyabhan007/linux-hpc-security): 105-chapter systems engineering curriculum — eBPF & observability, Ansible at scale, Slurm HPC & MPI, RDMA/InfiniBand fabric, Lustre/BeeGFS parallel storage, kernel tuning (NUMA, cgroups v2, hugepages), SOC/NOC operations, Splunk SIEM, chaos engineering, red/blue/purple teaming.

EDUCATION

Bachelor of Engineering (B.E.), Mechanical Engineering — First Class — Savitribai Phule Pune University — Dhole Patil College of Engineering 2014 (DPCOE)